



A multi-stack RNN-based neural machine translation model for English to Pakistan sign language translation

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Abstract

Sign languages are gesture-based languages used by the deaf community of the world. Every country has a different sign language and there are more than 200 sign languages in the world. American Sign Language (ASL), British Sign Language (BSL), and German Sign Language (DGS) are well-studied sign languages. Due to their different grammatical structure and word order the deaf people feel difficulty in reading and understanding the written text in natural languages. In order to enhance the cognitive ability of the deaf subjects some translation models have been developed for these languages that translate natural language text into corresponding sign language gestures. Most of the earlier natural to sign language translation models rely on rule-based approaches. Recently, some neural machine translation models have been proposed for ASL, BSL, DGS, and Arabic sign language. However, most of these models have low accuracy scores. This research provides an improved and novel multi-stack RNN-based neural machine translation model for natural to sign language translation. The proposed model is based on encoder–decoder architecture and incorporates attention mechanism and embeddings to improve the quality of results. Rigorous experimentation has been performed to compare the proposed multi-stack RNN-based model with baseline models. The experiments have been conducted using a sizeable translation corpus comprising of nearly 50,000 sentences for Pakistan Sign Language (PSL). The performance of the proposed neural machine translation model for PSL has been evaluated with the help of well-established evaluation measures including Bilingual Evaluation Understudy Score (BLEU), and Word Error Rate (WER). The results show that multi-stacked gated recurrent unit-based RNN model that employs Bahdanau attention mechanism and GloVe embedding performed the best showing the BLEU score of 0.83 and WER 0.17, which outperform the existing translation models. The proposed model has been exposed through a software system that converts the translated sentences into PSL gestures using an avatar. The evaluation of the usability has also been performed to see how effectively the avatar-based output helps compensating the cognitive hearing deficit for the deaf people. The results show that it works well for different granularity levels.

Keywords Pakistan sign language · Deaf people · Neural machine translation · Cognition · Avatar

1 Introduction

More than 5% of the world's population, that is almost 466 million, is affected by hearing disability. The projections show that by 2050 this number may rise to 900 million people [1]. Special languages, generally referred to as sign languages, are used by the deaf community of the world. Just like the natural languages there are different sign languages for different countries [2]. Each gesture comprises of some hand shapes, hand movements around the upper body, and facial expression including shoulder shrugging, lips movement, smile etc [3]. The grammatical structure of the sign language sentences is different from

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that of scriptive natural languages. Furthermore, the length of sentences in sign languages tends to be small [4]. Therefore, the deaf people face difficulty in reading and understanding most of the text available on the Internet, and in the books.

A diversified technological support has been provided to the deaf community to help them communicate with other people. This mainly revolves around the gesture recognition and natural to sign language translation tasks. The focus of this research is to address the problem of natural language text to sign language translation. The literature review reveals that most of the early machine translation models for natural to sign language translation are based on rule-based approaches [4, 5] and statistical approaches [6]. However, recently some neural machine translation models have been proposed for natural to American Sign Language (ASL) [7, 8], British Sign Language (BSL) [9], German Sign Language (DGS) [10, 11], and Arabic Sign Language (ArSL) [12].

Among these recent neural machine translation-based models for natural to sign language translation is a transformer-based translation model [13] for English to ASL which has been proposed by Manzano [8]. While, another model [7] presents a customization of IBM word alignment model for translating English to ASL. However, both models report a low BLEU score which is less than 0.20. While, a more recent neural machine translation model for Arabic to ArSL uses sentence morphology for encoding the sentences, and adopts traditional feed-forward and back-propagation neural network for the generation of translated sentence [12].

It can be observed that there exist language specific translation models for natural to sign language translations. As a matter of fact, every sign language has a different grammatical and morphological structure. Thus, there is a need to develop a specialized model for the translation of different sign languages. Particularly for PSL, there exists a rule-based machine translation model for natural language to PSL translation [14]. To the best of our knowledge, there is no neural machine translation model for the translating natural language input into equivalent PSL. Hence, there is a need to explore the potential of neural machine translation models for translating natural language to PSL.

It is worth mentioning that RNN-based encoder–decoder models have been found useful in conventional machine translation problems [15–18]. Hence, there is a need to explore their potential further for natural to sign language translation. To this end, this research presents a multi-stacked RNN-based encoder-decoder translation model that incorporates widely used attention mechanisms [15, 19, 20] to generate more accurate translation of natural language sentences into equivalent sign language sentences. Subsequently, the translated sentence is converted into avatar-

based output using word-level machine-readable repository of gestures. It uses a translation corpus of PSL comprising of 50,000 sentence pairs developed for training and testing machine translation models [21]. Further, a comparative analysis of the proposed multi-stack RNN-based encoder-decoder neural machine translation model, customized transfer learning model [8], and an existing rule-based translation model for PSL [14] has been presented in this research. The results show a BLEU score of 0.83 achieved by the proposed RNN-based model which outperforms the other two translation models for PSL.

The rest of the article has been structured in the following manner. Section 2 presents the details about the relevant literature. Details of the proposed RNN-based neural machine translation model for English to PSL translation have been presented in Sect. 3. The experimental setup and obtained results have been discussed in Sect. 4. A case study application that converts the translated output to an avatar has been presented in Sect. 5. Lastly, the article has been concluded in Sect. 6.

2 Related work

Sign language translation is a well-established branch of machine translation. The problem of sign language translation is a multi-disciplinary problem [4, 22]. It involves gesture recognition [23–28], natural to sign language translation [5, 29], repositories for sign languages [21–30], and generation of gestures with avatar technology [31–34]. It also involves specialized evaluation measures used to – machine translation problems [4, 35–37]. This research typically focuses on natural to sign language translation.

Many different translation models have been proposed for natural to sign language translation for different sign languages. This research started with some interesting research projects have also been conducted to address different aspects of natural to sign language translation for many English-based sign languages. These projects include TESSA [38], ZARDOZ [39], ASL Workbench [40], ViSiCAST [41]. Later on, for some other sign languages specialized domain-specific translation models have been proposed; for example South-African [42], Italian [43], and German [44] sign languages.

A careful analysis of natural to sign language translation research shows that initially rule-based translation models were proposed for solving this problem [45–47]. Later on with the development of language resources [48, 49] statistical machine translation models were also developed [50]. Some recent developments in this area show that many models have been proposed for some non-English, African, and Asian languages as well. Most of these are based on rule-based translation models including Arabic

Sign Language [47, 51–53], Greek Sign Language [54], Spanish Sign Language [55–57], South African Sign Language [42], Vietnamese sign language [58], Indian Sign Language [59], Bangla Sign Language [60], Thai Sign Language [61], and [14] for Pakistan Sign Language.

Recently, Neural Machine Translation (NMT) approaches have been used to solve the problem of natural to sign language translation problem [62]. These NMT approaches for sign language translation can be categorized into two different groups. First one focuses on text to text translation which is then converted into sign language gestures. While, the second one directly translates text into equivalent gestures. Table 1 presents a summary of these approaches.

It can be observed from the models presented in Table 1 that a couple of efforts have been made to translate English to ASL. Where, Cate and Hussain [7] adjusted IBM's word alignment model to propose English to ASL translation model. Later on, Manzano used transfer learning for English to ASL translation [8], which was trained and tested on the ASLG-PC dataset [122] that consists of 83,618 pairs of sentences. However, both these approaches have a low BLEU score of 0.17 and 0.18, respectively. Another effort for Arabic to ArSL machine translation has been carried out where it started with conventional rule-based approach [63], and later on was improved with the help of neural machine translation model that is referred to as ALTLA-SLang NMT [12]. The proposed model starts with pre-processing of data with morpho-syntactic analysis, whereby all the words in the sentence are encoded with morphological characteristics for encoding purpose. While, a feed-forward neural network with back-propagation is used to generate the target sentence of ArSL. A data set comprising of 9715 sentence pairs that translate Arabic to ArSL has been used for this purpose. The BLEU score reported for this work is 0.78. However, the test set for this

study was less than 100 sentences, which challenges the general applicability of this work. The afore-mentioned approaches use sentence pairs for training the model, while they convert the translated text into visual representation of gestures using avatar developed with the help of machine readable gestures' repository for their respective sign languages.

The other category of approaches for natural to sign language translation use data sets in which the natural language text is mapped onto the sign language gestures using text to gloss-based data. For instance, Mocilov et al. used transfer learning a variant of deep learning approach for translation of English into BSL [9]. This work used a nominal data set of 810 sentences of very small length. Two other neural machine translation models for translating German Sign Language (DGS) text to DGS Gloss using 8257 sequences of weather forecast videos. Whereby, Stoll et al. presented a deep learning model based on generative adversarial networks and motion generation with a BLEU score of 0.50 [10]. While, Saunders et al. combined RNN with motion graph to convert German into DGS gloss with a BLEU score of 0.46 [11]. These approaches involve different type of data sets which map spoken or written text directly onto the video gestures. Hence, these approaches are fundamentally different from the work being presented in this research.

It is imperative to mention that the Recurrent Neural Network (RNN)-based sequence-to-sequence architectures have been found very successful in performing language translation tasks [15–17]. These approaches use encoder and decoder, both of which comprise of an RNN. In order to handle longer sequences Long Short-Term Memory (LSTM) [64] and Gated Recurrent Units (GRU) [18] have been used as RNN cells. Such architectures were further improved with the introduction of attention mechanisms

Table 1 Neural machine translation models for natural to sign language

Language	Category	Description	Data set	Accuracy (BLEU)
English to ASL [7]	Text to Text to Sign	Adjustment to IBM word-alignment model	579 sentence pairs	0.17
English to ASL [8]	Text to Text to Sign	Transfer learning-based model	87,000 sentence pairs	0.18
Arabic to ArSL [63]	Text to Text to Sign	Feed-forward ANN-based Encoder-Decoder model because of small data size	9715 pairs of sentences	0.79
BSL to Gloss [9]	Text to Sign	Transfer learning-based English to BSL (video mapping)	810 sentences of very small length	Not reported
DGS to Gloss [11]	Text to Sign	RNN Combined with motion graph	8257 sequence of weather forecast videos	0.46
DGS to Gloss [10]	Text to Sign	Progressive transformer with adversarial discriminator	8257 sequence of weather forecast videos	BLEU 0.50

[15, 19, 20] were introduced to handle the longer sequences. Therefore, this research presents an improved multi-stack RNN-based encoder-decoder model that involves attention mechanism for translating natural language to sign language and has been tested for Pakistan Sign Language.

Another important aspect of machine translation models is the provision of translation data sets that help in training and testing the models. A list of some prominent data sources used for developing different natural to sign language translation models for some sign languages has been presented in Table 2. It can be observed that in most of the cases, the number of sentences in these data sets is very small. Though, modern data-driven machine translation approaches require large amount of data for better results. It is imperative to mention that significantly larger data sets are available for speech recognition and generation [4]. Hence, there is a need to develop new sizeable datasets and extend the existing data sets for developing improved translation models.

Pakistan Sign Language is not very well-studied sign language. Creation of Pakistan Sign Language resources and technologies is in early stages [3]. The work began in early 2000 [65, 66] with a study that attempted to collect data for words and phrases in PSL. Following that, however, no significant effort has been made to automate the translation of natural language into PSL. Some recent efforts have been made to layout an architectural platform to convert natural language text into PSL, and vice versa [3, 67, 68]. Later on, a grammar-based machine translation model has been proposed and tested on a small data corpus [14]. Similarly, another useful effort has been made to develop a sizeable translation corpus for PSL, comprising

of nearly 50,000 sentences [21]. This invites the researchers to evaluate the effectiveness of modern neural machine translation models for English to PSL translation. Hence, this research presents an RNN-based neural machine translation model for translating English text into equivalent Pakistan Sign Language sentences using the PSL translation corpus [21].

3 Methodology

The proposed multi-stacked RNN-based neural machine translation model that takes English language sentence as input and translates it to equivalent PSL sentence has been discussed in this section. A high level diagram of the proposed model has been presented in Fig. 1. This involves some preprocessing on raw data. Then the model takes input sentences and processes them with the help of encoder. The encoder generates an intermediate representation using attention mechanism that helps to emphasize words while considering the context of the sentence. The intermediate representation is then passed to the decoder that generates the final translation. The decoder also uses attention mechanism to produce the output. This section presents the details of all these steps.

3.1 Details of the proposed RNN-based model

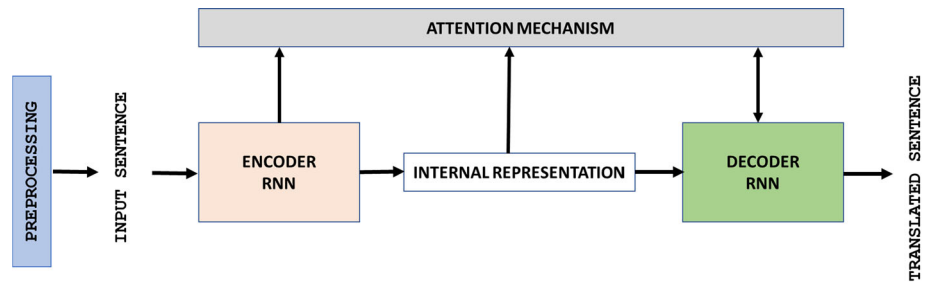
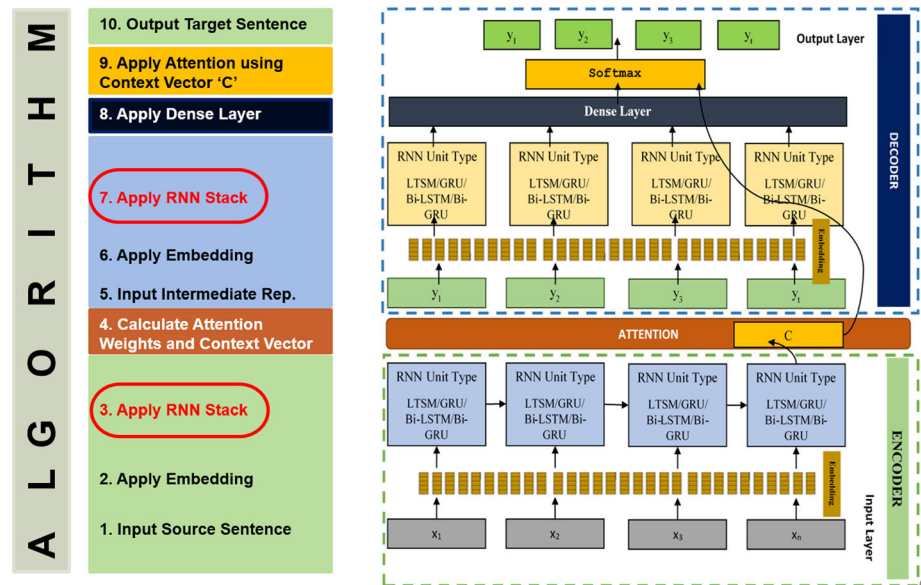
The detailed architecture of the proposed multi-stacked RNN model for natural to sign language translation has been presented in Fig. 2. It uses encoder-decoder architecture that employs attention mechanism and embeddings. It is imperative to note that the proposed architecture comprises of multi-stack RNN in encoder and decoder. It is important to note that every machine translation problem requires a distinct configuration of such encoder-decoder model. Algorithm 1 discusses how the input is processed through the encoder/decoder architecture. This section discusses all the components of the proposed model, and in the end presents some details of the optimal parameter settings of the proposed model for English to PSL translation.

3.2 Rationale of the proposed multi-stack RNN model

The proposed model enhances the tradition encoder decoder model by stacking multiple layers of RNN, where each layer contains multiple memory cells. Figure 2 presents the architecture of the proposed multi-stack deep learning model used for natural to sign language translation. The depth offered by multiple stack layers of RNN allows the hidden state at each level to operate at different timescales.

Table 2 Corpora for natural to sign language [55]

Corpus	Language pairs	Sentences
Othman and Jemni	English-ASL	431
Bonham	English-ASL	11,000
Cate and Hussain	English-ASL	600
Manzano	English-ASL	83,618
–	Italian–Italian SL	585
RWTH-phoenix	German–German SL	2486
–	Chinese–Taiwanese SL	1983
–	Catalan–Catalan SL	199
ATIS	English-iris SL	595
ATIS	German–German SL	595
ATIS	English south African SL	595
ID	Spanish–Spanish SL	2000
DL	Spanish–Spanish SL	2000
eSign-3D	Spanish–Spanish SL	5000

Fig. 1 High-level neural machine translation model**Fig. 2** Proposed RNN-based deep learning model

This stacking is analogous to incorporating multiple feed-forward layers on top of each other in conventional deep neural networks. However, in the case of an RNN stack layer, the upper layer provides a sequence of output instead of a single value to the layer below it. That is, a specific output for each input time-step instead of one output, for all the input time-steps. This improvisation helps the model capturing the syntactic and semantic information in a better way which helps to improve accuracy.

Furthermore, this multi-stack RNN model utilizes the embeddings and attention mechanisms in processing the input sequence in the encoder to generate a context vector. This context vector is then used to generate the output sequence more accurately. Hence, the proposed model is a multi-stack RNN-based encoder-decoder neural machine translation model which also employs embeddings and attention mechanisms. Thus, it is an improvised neural machine translation model that addresses the natural sign language translation problem. Algorithm 1 shows how the input is processed through different layers of the proposed model.

Algorithm 1

1. Firstly the input sentence is word-by-word fed to the encoder
2. Apply embedding to it to convert it into fixed sized vector.
3. Apply multi-stack RNN to the embedded vectors. (most suitable number is empirically found with experiments)
4. Attention vector and attention weights are calculated. While the context vector is calculated using the average sum of attention vector values at each time step.
5. The output of the last hidden layer of the encoder along with the context vector is fed to the decoder.
6. The decoder again processes this intermediate representation of the sentence using embeddings
7. Apply multi-stacked layers of RNN
8. Process through the dense layer.
9. Use the context vector 'C' to generate the target sentence in correct word order.
10. Output the translated sequence of words

3.3 Encoder

The encoder takes the input text and starts processing it with the help of embedding layer which converts the input word into fixed sized vector. This vector is passed to the RNN unit used in the encoder. The RNN unit produces an intermediate representation with the help of attention mechanism used. The details of these components have been discussed below.

Embedding layer: The preprocessed input text is passed to the model through the encoder, which starts processing it using the embedding layer. The text cannot be directly fed for training to the machine learning or deep learning models, because machine learning algorithms and deep learning architectures are unable to process strings in raw form, as they require numbers as inputs to perform any task. The embedding layer helps to convert each word into fixed sized vector. The proposed translation model understands and performs operation on this textual data with its representation in the form of fixed sized vector by using pre-trained embeddings. This research uses pre-trained embeddings GloVe and Fasttext. After empirical evaluation through experimentation the dimensions of the embedding have been set to 300. The vectorized output is then passed on to the RNN hidden layers.

Recurrent neural network (RNN) units: The RNN units have been used in Encoder and Decoder: Recurrent neural network are a type of neural network which are suitable for modeling sequence-sequence data like text. RNN models have two main variants. Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU). Both of them have bi-directional variants called Bi-LSTM and Bi-GRU. Thus, these different variants of RNNs have been used for the experimentation purpose to empirically find the best performing variant.

The encoder processes each word of input sequence step-by-step and produces output that is used by the next time step. The final output of the encoder is termed as the context vector which is then passed on to the decoder for further processing.

In this research, the encoder is a set of multiple recurrent cells either GRUs or LSTMs or their bidirectional variants, where each cell takes one word as an input x_i and updates the hidden state h_i . After reading all the inputs, the encoder generates the final hidden vector h_t that summarizes the information for all input words, and then the decoder takes that final hidden vector and generates output words one by one. The hidden state h_t is computed using the given formula:

$$h_t = f(W^{(hh)}h_{t-1})$$

The formula represents the result of a normal recurrent neural network. Whereby, the model applies suitable weights W to the previous hidden state h_{t-1} and the input vector x_t .

3.4 Decoder

Decoder is also a stack of recurrent units which can be LSTMs or GRUs or their bi-directional variants. Decoder generates the output sequence by predicting the next output y_t at timestamp t given the hidden state h_t . The input of the decoder is the final hidden vector that summarizes the whole input sequence obtained at the end of the encoder model. Any hidden state h_t at each timestamp is computed using the following formula:

$$y_t = \text{softmax}(W^{(s)}h_t)$$

The model calculates the outputs using the hidden state at the current timestamp together with the corresponding weight matrix $W^{(s)}$. Probability vector is created using softmax that assists it to determine the output word. In short, the decoder processes the intermediate representation to produce the final output. In the whole process the attention mechanism serves as the method to align and translate. The decoder uses a similar architecture including embedding layer, followed by RNN hidden layers, which passes on the output to the dense layer. While, dropout and batch normalization were also used after fully connected dense layer to avoid over-fitting, followed by *softmax* activation function before the final output. Thereafter the final output is returned in the form of sequence of target language words.

3.5 Attention mechanism

RNN cannot recall longer sentences because of vanishing or exploding gradient problems. It can only remember the parts which it has recently gone through. In [15] the research proposed the encoder-decoder network, and exhibited that the performance of the encoder-decoder network reduces as the length of the sequence increases.

Even though an RNN is supposed to capture the long-term dependency, it will in general get distracted in explicit cases. Another issue is that RNNs do not have any capability to give more significance to a portion of the information contrasted with others while translating the sentence. The attention mechanism helps the model aligning the translated sentence by focusing on specific parts of the input sequence when predicting the corresponding output sequence. Hence, the attention mechanism enables better quality learning and more appropriate output. This research employs some frequently used attention

mechanism introduced by Bahdanau [15, 19] and Luong [20].

3.6 Parameter settings for the proposed translation model

Extensive experimental evaluation has been performed to figure out the optimal parameter settings for the proposed model to translate from English to PSL over the corpus of 50,000 sentences. A number of experiments were designed and executed based on the variations of all four variants of RNNs as encoder and decoder; in terms of embedding GloVe and Fastext; whereas Bahdanau and Luong were used as possible attention mechanism. Apart from this, some other parameters have also been optimized that include batch size and loss function. More details about the experimental setup and obtained results have been presented in Sect. 4.

4 Experimental evaluation

4.1 Experimental settings

This section presents the details of the deep learning models used for conducting the experiments. In order to compare the results of different models, it is imperative to conduct the experiments on the same data set. Thus it also explains about the data set used for performing experiments. Lastly, it also presents the specialized measures used to evaluate and compare the efficacy of the translation models.

Machine translation models used for experiments: This section presents the evaluation and comparative analysis of the aforementioned recurrent neural network-based models for natural to Pakistan Sign Language Translation. A set of experiments has been conducted using the proposed RNN-based encoder-decoder model trained by using both simple and bidirectional LSTM and GRU along with pre-trained embeddings namely Fastext and GloVe. Where, widely used attention mechanisms have also been used to improve the quality of results. The implementation has been conducted using Keras library.

While, the second set of experiments has been conducted to empirically evaluate the efficacy of transfer learning model by customizing an existing model proposed for English to ASL translation [8] for translating English to PSL with different experimental settings. Lastly, the results of both these models have been compared with an existing rule-based translation model for translating English into PSL [14].

Thus, the following three types of machine translation approaches have been used for the evaluation and comparative analysis.

- (i) Baseline RNN Model
- (ii) Proposed RNN and Attention-based Machine Translation Model
- (iii) Customized Transfer Learning Model (Customization of [8])
- (iv) Rule-Based Translation Model for English to PSL Translation [14]

Data set used for experiments: The data set generated using crowdsourcing platform that comprises of 50,000 sentences has been used to run these experiments. Whereby, training data set comprises of 42,000 sentences, while test data set comprises of 8000 sentences.

Evaluation measures: The results of these experiments have been assessed using famous evaluation measures BiLingual Evaluation Understudy score (BLEU) proposed by Papineni [69], and Word Error Rate (WER) proposed by McCowan [29].

The BLEU score is based on the n-gram overlapping between the translation output and corresponding reference sentence. This is computed and reported for different n-grams, in general BLEU-1 to BLEU-4. This research also reports the results up to BLEU-4. While the Word Error Rate is a Levenshtein distance between translated output and the reference sentence.

4.2 Experiments with proposed RNN-based encoder-decoder and attention model

Experiments have been conducted to find the optimal settings for RNN-based encoder-decoder model for machine translation from English to PSL. The operating parameters for executing a set of experiments to figure out optimal parameter settings for the proposed deep learning translation model have been presented in Table 3.

GRU, LSTM, Bi-GRU, and Bi-LSTM are different types of RNN models have been used in encoder and decoder. Whereas, GloVe and Fastext have been used a pre-trained embeddings for the experimentation. Similarly, Bahdanau and Luong are the attention mechanisms which

Table 3 Main operating parameters to design experiments

Parameters	Values/options
Type of RNN	GRU, LSTM, BiGRU, BiLSTM
Embedding	GloVe, Fastext
Attention mechanism	Bahdanau, Luong

have been widely used in the literature related to the machine translation problems. This results into 4 different experiments for each type of RNN, thus resulting into 16 total experiments.

The other parameters used for tuning the models have been presented in Table 4. This includes different possible values for batch size, number of units in hidden layers, and loss function were used as used in many other studies in the literature [5, 8, 10, 12]. For each experiment Grid search has been used by the Keras optimizer to empirically find the best parameter settings for all different settings of experiments. In all experiments the choices of the number of RNN layers and the number of units in dense layers are set empirically. In all experiments Adam optimizer has been considered suitable because it has been reported to be the better optimizer for machine translation problems [5, 8, 10]. The model is trained using 30 epochs, and the training is stopped if there is no decrease in the validation error for 10 epochs.

Table 5 shows the results obtained from the experiments conducted with baseline models for different variants of RNN in the encoder and decoder. The results show that GRU performs better than the other variants of RNN.

Table 6 shows the results obtained from the experiments conducted with different configurations of RNN-based encoder-decoder deep learning model with different attention mechanisms, and pre-defined embeddings.

Best configuration of the translation model: It can be observed from Table 6 that experiments conducted using Bahdanau attention and GloVe embedding outperform the rest of the variants for every RNN variant. While, GRU encoder-decoder with GloVe embedding and Bahdanau attention stands out as the best model. In all experiment categorical cross entropy was found to be a better loss

Table 5 Comparison of baseline RNN models

	BLEU-1	BLEU-2	BLEU-3	BLEU-4	WER
Experiments with baseline RNN					
Encoder/decoder					
GRU	0.6132	0.5064	0.4252	0.3603	0.39
LSTM	0.5892	0.4707	0.3892	0.3246	0.42
BiGRU	0.6016	0.4934	0.4161	0.3443	0.41
BiLSTM	0.5735	0.4787	0.3835	0.3207	0.43

Best results in bold

function over binary cross entropy. Whereas, for optimizer, “Adam” was used for all experiments. It is imperative to note that Adam optimizer dynamically adjusts the learning rate, with a larger step size in the beginning, and small values to for slow learning in the later stages of training. Whereas, the dropout rate was chosen from [0.1, 0.2, 0.3, 0.4, 0.5], where the 0.2 was chosen as the best value. The number of units for each layer were 1024. While, the batch size of 32 was chosen to be the optimal from [16, 32, 64, 128]. It is important to note that for the sake of clarity the rest of the parameter settings have been presented for the best experimental setup, that is GRU with GloVe embedding and Bahdanau attention mechanism.

The empirical results show that the combination of GRU (RNN), GloVe (embedding), and Bahdanau (Attention) turned out to be the best combination. Though the deep learning models work like a black-box [70], yet the findings can be rationalized in the following manner. GRU has been found to be more effective as compared to LSTM. The main reason is small length of sentences, as LSTM works well for longer sequences, which GRU is found to be more suitable for short sequences, and the average length

Table 4 Parameters used and tuned by the optimizer

Parameters	Values/options
Encoder layers	1
Decoder layers	1
Encoder hidden units	256, 512, 1024
Decoder hidden units	256, 512, 1024
Optimizer	Adam
Batch size	16, 32 , 64, 128
Dropout	[0.1, 0.2 , 0.3, 0.4, 0.5]
Activation function	Sigmoid
Loss function	Categorical Cross Entropy , Binary Cross Entropy
Maximum input length	20
Maximum target length	18
Parameter tuning method	Grid search
Epochs	30
Best results in bold	

Table 6 Comparison of RNN variants with different attention mechanisms and embedding

	BLEU-I	BLEU-2	BLEU-3	BLEU-4	WER
<i>Encoder/decoder GRU</i>					
Luong + Fasttext	0.7671	0.6251	0.5231	0.4463	0.24
Luong + GloVe	0.7948	0.6516	0.5412	0.4618	0.21
Bahdanau + Fasttext	0.781	0.6464	0.5314	0.4546	0.22
Bahdanau + GloVe	0.8139	0.6759	0.5697	0.4769	0.19
<i>Encoder/decoder LSTM</i>					
Luong + Fasttext	0.6452	0.5115	0.4193	0.3423	0.36
Luong + GloVe	0.6081	0.4971	0.3954	0.3156	0.41
Bahdanau + Fasttext	0.6485	0.5267	0.4247	0.3442	0.35
Bahdanau + GloVe	0.6793	0.5383	0.4366	0.3574	0.34
<i>Encoder/decoder BiGRU</i>					
Luong + Fasttext	0.7496	0.6128	0.5085	0.4117	0.26
Luong + GloVe	0.7531	0.6208	0.5136	0.4375	0.24
Bahdanau + Fasttext	0.7509	0.6227	0.5127	0.4286	0.24
Bahdanau + GloVe	0.7828	0.6412	0.5328	0.4528	0.22
<i>Encoder/decoder BiLSTM</i>					
Luong + Fasttext	0.6653	0.5517	0.4685	0.3954	0.34
Luong + GloVe	0.6952	0.5815	0.4802	0.4078	0.32
Bahdanau + Fasttext	0.6773	0.5473	0.4434	0.3751	0.33
Bahdanau + GloVe	0.7484	0.6011	0.5022	0.4159	0.26

Best results in bold

of the sentences is 8 words. A moderate batch size of 32 has been found to be more suitable for the model. While among the embeddings GloVe performed better as compared to FastText as it performs well on small data set. On the other hand, Bahndanau attention performed better as compared to Luong as it uses global attention model and depends on previous hidden state and uses additive attention.

Lastly, experiments with multi-stack RNN-based encoder-decoder model have conducted to figure out the most suitable number of stacked RNN layers. Since, GRU performed better than the other variants of RNN, so it was used to experiment for different stacked layers of RNN. While, the other parameters for the best configuration were

retained for these experiments. Hence, Bahdanau attention, and GloVe embedding were used for all these experiments.

The obtained results have been presented in Table 7. The results show that 2 stacked RNN performs better than the other values. It tends to over-fit with the higher number of stacked layers of RNN.

4.3 Experiments with transformer model

This section presents the results obtained from the transformer-based translation model on the data set developed for PSL. This research customizes the transformer model proposed in [8] for translation of English to PSL.

Model size: The transformer is a stack of encoder and decoder and the count of these encoder and decoder layer N_{Layer} is a hyper parameter that we can test our model on. Generally, Transformer use multiple encoder-decoder layers, and its tuning involves trying different parameter values to solve their problem. So, it is obvious to test different values for N_{Layer} for the English to PSL translation task. The original paper reports 6 layers. While, in this experiment the performance of 1–6 encoder-decoder layers has been empirically evaluated to figure out the best number of encoder decoder layers. The original paper uses batch size of 64, but in case of PSL the batch size of 32 performed better than that of 64. The results have been reported in Table 8, and it can be observed that with the

Table 7 Experiments for GRU with different stacked layers of RNN

	BLEU-I	BLEU-2	BLEU-3	BLEU-4	WER
Experiments with GRU					
Encoder/decoder					
1	0.8139	0.6759	0.5697	0.4769	0.19
2	0.8327	0.7001	0.6093	0.5141	0.17
3	0.7771	0.6382	0.5225	0.4413	0.23
4	0.7623	0.6298	0.5181	0.4238	0.24

Best results in bold

Table 8 Transformer: experiments on different number of encoder-decoder layers

Layers	BLEU-1	BLEU-2	BLEU-3	BLEU-4	WER
1	0.7033	0.5734	0.4619	0.3734	0.31
2	0.7521	0.6309	0.5259	0.4321	0.25
3	0.7413	0.6182	0.5123	0.4224	0.26
4	0.7111	0.5914	0.4914	0.4137	0.29
5	0.6832	0.5858	0.4664	0.3819	0.32
6	0.6453	0.5228	0.4337	0.3717	0.36

Best results in bold

model starts over-fitting on the test set with the increase in the number of layers. The table shows that the model with 2 encoders and decoders outperforms on both test and validation data. A negligible difference has been observed while comparing the results of evaluation matrices of the model with layers 2 and 3. However, it is obvious that the use of 2 layers is computationally inexpensive as compared to using 3 layers, so it is recommended to use 2 layers for English to PSL translation task. Generally, such models perform better on a higher count of encoder-decoder layers, but in this case the problem eventually boils down to English to PSL (English) translation, which is potentially less complex as compared to totally different languages like English to French.

4.4 Comparison of proposed work with state-of-the-art models

This section presents a comparison of transformer, RNN, and Grammar-based approach. The comparison involves best performing RNN, customized transformer model [8] implemented in this research, and grammar-based approach proposed by [14].

Table 9 and Fig. 3 show the comparison of all approaches and it can be observed that the best variant of RNN-based encoder-decoder outperforms that transformer model and the grammar-based translation model for PSL. It is imperative to mention that embeddings imparts semantic richness to the RNN model thus helping it dealing with out-of-vocabulary words in a useful manner. While, the attention mechanism enables it to perform better word-

alignment using the context vector, resulting in better BLEU score. Generally, the transformer models tend to performs well on large sentences, but in this study the average length of a sentence is almost 8 words, which is not considered to be a long sentence. Since from the literature it has been observed that sign language sentences are short in order to understand the deaf people [10, 22]. Therefore, both RNN exhibits better results as compared to the transformer model. As far as the grammar-based machine translation model is concerned, it is strictly based on the defined rules, and may not perform well on larger data sets with different categories of sentences.

Furthermore, it will be interesting to extend the data set by including a significant number of more sentences, particularly by including longer sentences to see the comparison of the two approaches.

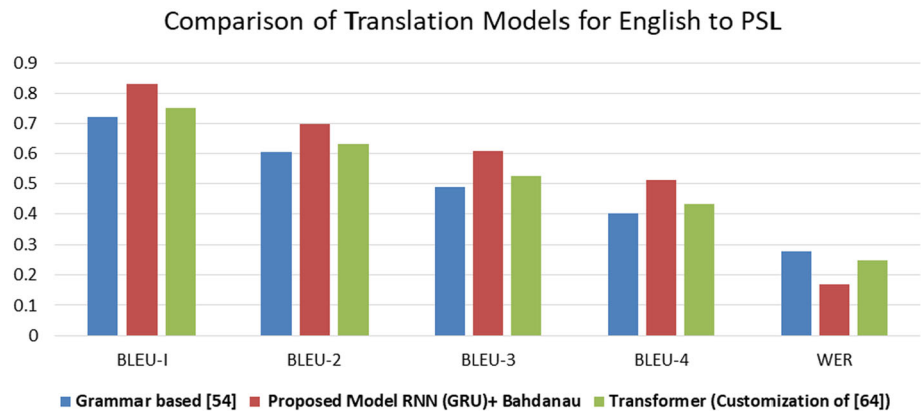
4.5 Qualitative analysis

This section presents the qualitative results by showing the sample output of the best RNN-based translation model and transformer model used in the experiments. It is a simple and widely used qualitative method to inspect and compare the resultant translations. Table 10 presents some of the translations generated from the proposed RNN-based and customizer transformer model along with ground truth translations. Although the named entities like person names and locations are limited in our data yet the RNN-based model seems to perform better at capturing these details it shows that it learned the embeddings well. In some of the translations exact wording is different but it passes on similar information. Explicit numbers are likewise difficult as there is no linguistic setting to recognize one from another. As a whole, with these settings both RNN-based model and transformer model have performed reasonably well, though the RNN-based model performed better as compared to the transformer model.

Table 9 Comparison of different machine translation approaches

	BLEU-1	BLEU-2	BLEU-3	BLEU-4	WER
Grammar based [14]	0.7221	0.6068	0.4902	0.4012	0.28
Proposed model RNN (GRU) + Bahdanau	0.8327	0.7001	0.6093	0.5141	0.17
Transformer (customization of [8])	0.7521	0.6309	0.5259	0.4321	0.25

Best results in bold

Fig. 3 Comparison of transformer and RNN**Table 10** Translations generated by RNN and transformer

Ground truth:	Was i jacket buy
RNN:	Was i jacket buy
Transformer:	Was i jacket buy
Ground truth:	John arithmetic like
RNN:	John arithmetic like
Transformer:	John moments like
Ground truth:	Was Aliyan say was he know Javeria children
RNN:	Was Aliyan say was he know Javeria children
Transformer:	Was Aliyan say was he know Javeria children
Ground truth:	Was he suddenly door close
RNN:	Was he suddenly door close
Transformer:	Was he door suddenly close
Ground truth:	Everybody once or twice fail
RNN:	Everybody once or twice fail
Transformer:	Everybody once twice fail
Ground truth:	Was she as child to Lahore come
RNN:	Was she child to Lahore come
Transformer:	Was she as child to Lahore come

5 English to PSL translation prototype application

The translation system has been exposed in the form of a web application that allows the users to input English language sentences. These sentences are then translated by the proposed translation neural machine translation model. Subsequently, the word level parallel corpus is used to build and play the avatar that performs the gestures for Pakistan Sign Language. Figure 4 presents a snapshot of the web application developed for natural language to PSL translation. Whereas, Fig. 5 shows the avatar playing the translation by rendering the HamNoSys for the words used in the translated sentence.

5.1 Evaluation of the understandability of the translation system

A small scale evaluation for the understandability of the sentence and paragraph level translation has also been conducted for the neural machine translation model. Whereby, 30 students were involved to provide their feedback on the understandability of the output avatar generated by the web-based translation system. The users were asked to rate the degree to which they managed to understand the avatar on the scale of 1–10.

Almost 100 medium length sentences from difference tenses were used to evaluate the avatar-based output. Whereas, 5 paragraphs comprising of 5–8 sentences have been used for testing purpose. The average score for the understanding of the sentence level translation is 8.4. While, the average score for paragraphs is 8.1.

6 Conclusion

This work presents an improved multi-stack RNN model for natural to Pakistan sign language translation. The proposed model is based on the conventional encoder-decoder architecture that involves attention mechanism and pre-trained embeddings. It further enriches the baseline encoder decoder architecture with multi-stack RNN. The experimental evaluation for the proposed model was carried out using a translation data set of nearly 50,000 sentence pairs for Pakistan Sign Language, while employing well-regarded evaluation measures used for machine translation tasks which include, BLEU score and Word Error Rate (WER). Whereby, different variations of the operating parameters involving batch size, pre-trained embeddings, and attention mechanisms have been utilized to design a variety of experiments to find the most suitable configuration of the model.

The results reveal that GRU, with a combination of Bahdanau attention and GloVe embedding, is the most

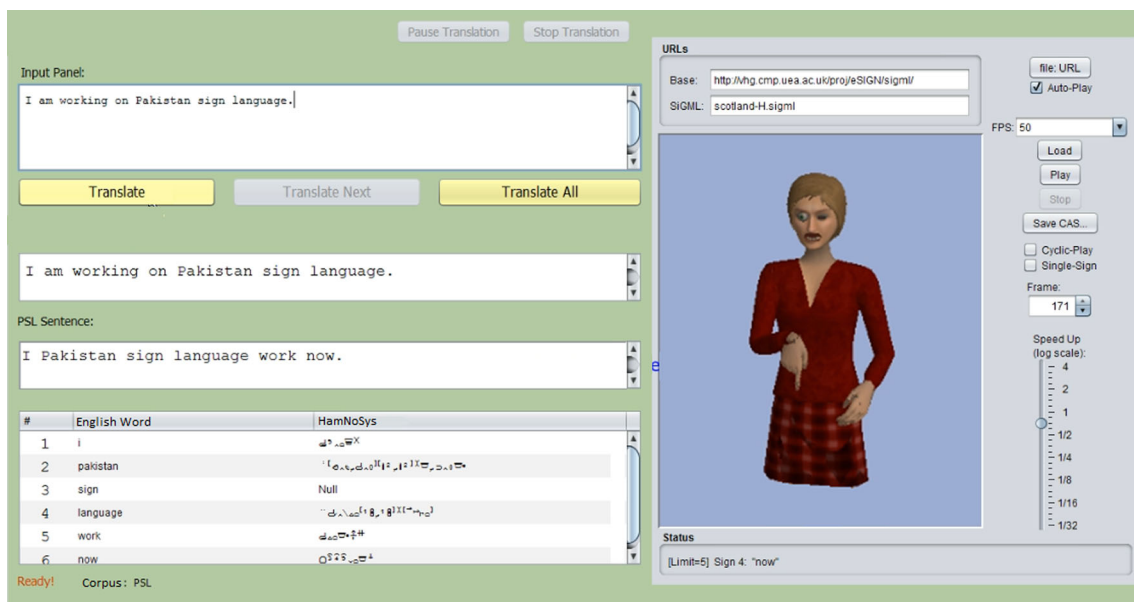
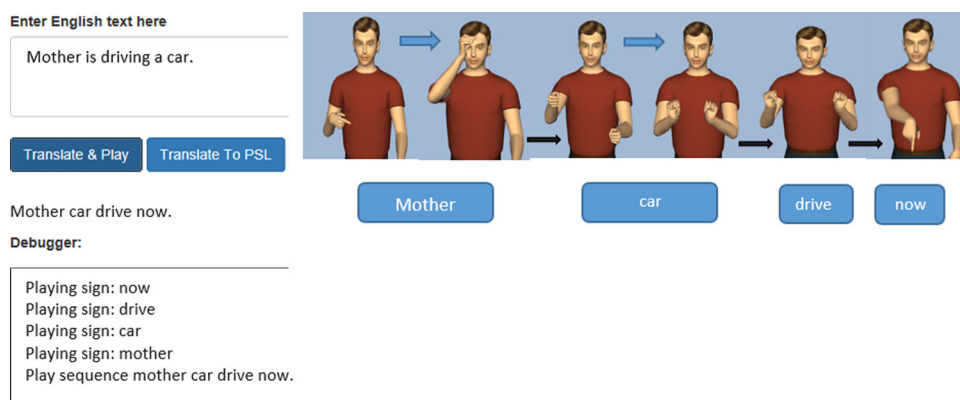


Fig. 4 Web interface for natural language to PSL translation

Fig. 5 Avatar for translated output



suitable RNN variant for this translation task. Furthermore, the number of GRU layers in multi-stack RNN architecture were empirically calculated, and the experimental evaluation reveals that 2 stacks of GRU performs better than the others.

In order to compare the proposed approach with existing relevant models, the results were compared with transformer model, and existing rule-based translation model. The results show that the proposed encoder-decoder-based multi-stacked RNN out-performed the customized transformer architecture, and the rule-based translation model by [14]. Where the proposed model attained the BLEU-1 score of 0.83, BLEU-4 score of 0.51, and WER score of 0.17.

This research work can be extended by incorporating sentences with larger length and unique words into the corpus. Similarly, generating English from PSL input that is the other way around is an even more interesting

challenge. Similarly, the model can be tested and customized for other sign languages as well.

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Declarations

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Code availability Not applicable.

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